

RECSYS CHALLENGE 2026 · MUSIC-CRS

# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color, creating a textured, rhythmic pattern.



# Contents

SIX PARTS

OPTIONAL

|          |                                  |                                                             |
|----------|----------------------------------|-------------------------------------------------------------|
| <b>1</b> | <b>The problem</b>               | finding music by talking, and why it is hard                |
| <b>2</b> | <b>Where the data comes from</b> | how the organisers built the conversations                  |
| <b>3</b> | <b>How it is scored</b>          | one right track per turn, plus a judged reply               |
| <b>4</b> | <b>Our approach</b>              | read the chat, search, rank, reply                          |
| <b>5</b> | <b>How it scored</b>             | the official results                                        |
| <b>6</b> | <b>What we learned</b>           | four lessons, each with its evidence                        |
| <b>A</b> | <b>Appendix</b>                  | every detail: schema, branches, ranker, audits, relabelling |



# 1 The problem

finding music by talking, and why it is hard



# Finding music by talking

THE PROBLEM

OPTIONAL

I'm trying to remember a song that feels like a quiet, personal story...



Neko Case — "Calling Cards"

Not quite the one... it definitely had a female singer.



Neko Case — "Man"

No, that's not it either. Much more subdued... almost a cappella.



Your system answers here: **20 tracks** and a **short reply**

A real conversation from the challenge's final test set.



# Why it is hard

THE PROBLEM

OPTIONAL



## It builds up

Each turn refines the last: a mood, then a singer, then “more subdued”.



## No means no

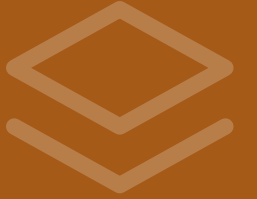
A track or artist the listener turned down should stay out.



# 47,071

## One right answer

tracks in the catalog, and one of them is counted as correct.



2

# Where the data comes from

how the organisers built the conversations





# The conversations are simulated

THE DATA

OPTIONAL



## A real listening session

from LFM-2b, 2005–2020



## Profile agent

writes a listener persona from 5 of their tracks



## Goal agent

picks 1 of 44 goals — e.g. “find a song by its lyrics”



## Listener and recommender

two agents talk for 8 turns

Every agent is **Gemini 2.5 Flash**. The conversations are written by LLMs, on top of real listening histories.

Source: TalkPlayData 2 (Doh et al., 2025), §3



# Who knows what

THE DATA

OPTIONAL



## Listener agent

- the listener's profile
- the goal

After each pick, marks it **MOVES** or **DOES NOT MOVE** toward the goal.



## Recommender agent

- the listener's profile
- a pool of 16–32 tracks
- ~~the goal~~

Must pick one track from that pool each turn.

The recommender agent's pick **is the answer key** the challenge scores you against.

Source: TalkPlayData 2 (Doh et al., 2025), §3.2



# What you get

THE DATA

OPTIONAL



**15,199**

**training  
conversations**

8 turns each, with the  
answer for every turn



**1,000**

**development  
conversations**

with answers — for  
checking yourself



**2 × 80**

**blind test  
conversations**

final turn only; scored  
on the leaderboard



**47,071**

**tracks in the  
catalog**

title, artist, tags, plus  
text, audio and cover  
embeddings



# 3 How it is scored

one right track per turn, plus a judged reply





# Points for where you rank it

SCORING

OPTIONAL

|        |  |      |
|--------|--|------|
| 1st    |  | 1.00 |
| 2nd    |  | 0.63 |
| 3rd    |  | 0.50 |
| 5th    |  | 0.39 |
| 10th   |  | 0.29 |
| 20th   |  | 0.23 |
| 21st + |  | 0.00 |

## NDCG@20

One correct track per turn. Points depend only on **where** you placed it in your 20.

## SO

Second place costs a third of the point.  
Outside the top 20 earns nothing.



# The final score

SCORING

OPTIONAL

50%

right track · nDCG@20

30%

reply quality · LLM judge, 1–5

10%

catalog  
diversity

10%

reply wording

## LLM JUDGE

The organisers' LLM reads the conversation and your reply, and rates it 1–5.

## CATALOG DIVERSITY

How many **different** tracks you recommend across all turns, as a share of the catalog.

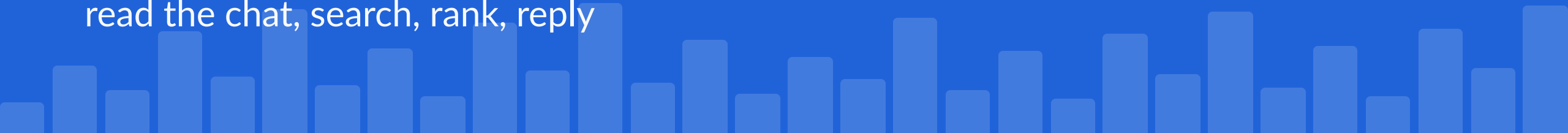
## REPLY WORDING

Lexical diversity: how varied the words in your replies are.



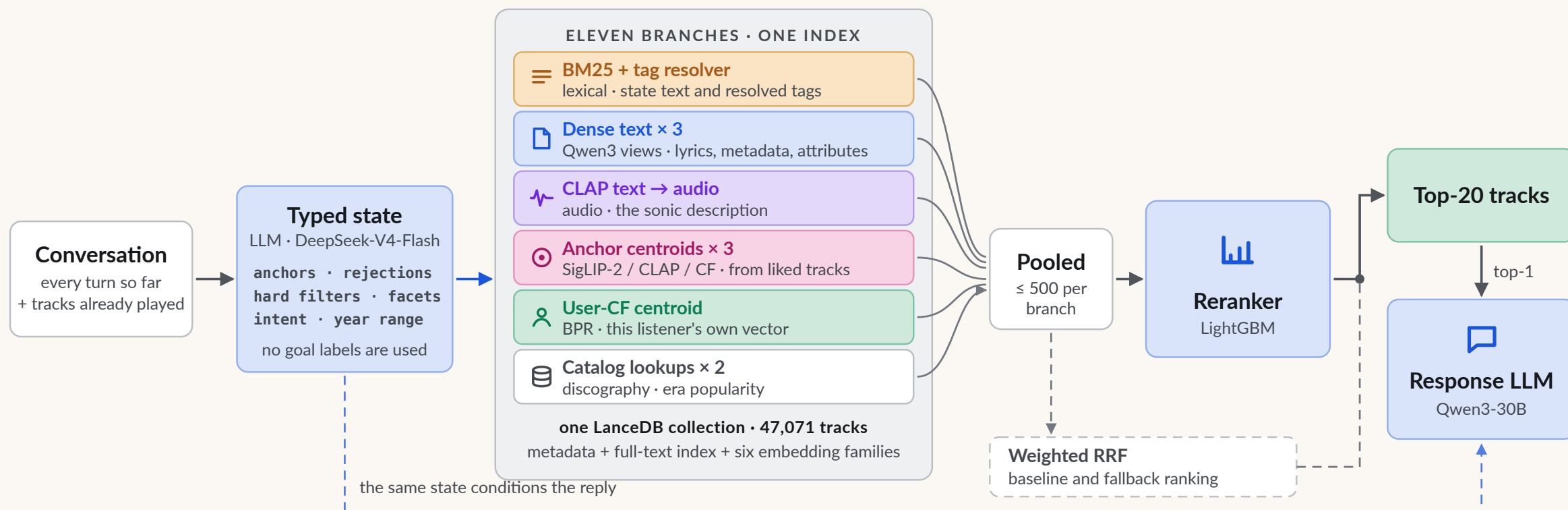
# 4 Our approach

read the chat, search, rank, reply



# ≡ One state, eleven branches, one learned ranker

APPROACH CORE



# Step 1 — read the chat into a state

APPROACH **CORE**

 “Calling Cards” — *not quite*

 “Man” — *not it either*

Much more subdued... a sense of place in the lyrics... almost a cappella.

LLM  
→

|             |                          |
|-------------|--------------------------|
| STAY WITH   | Neko Case                |
| TURNED DOWN | “Calling Cards”, “Man”   |
| MOOD        | subdued                  |
| LYRICS      | a sense of place         |
| SOUND       | stark, almost a cappella |

Every later step reads these fields, not the raw conversation.

## Step 2 – search eleven ways

APPROACH

OPTIONAL



×1

### Words

matching words in titles and tags



×3

### Meaning

similar descriptions, lyrics and metadata



×1

### Sound

what the track sounds like



×3

### Like these

near the tracks the listener liked



×1

### This listener

their own listening history



×2

### Lookups

the artist's discography, hits from an era

---

Up to **500 candidates** from each search, pooled into one list.



## Step 3 – rank the pool

APPROACH

OPTIONAL



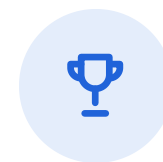
**The pooled candidates**

up to 500 from each search



**LightGBM ranker**

scores each on 146 signals



**Top 20**

the list that gets scored

---

Its strongest signal: a model we **fine-tuned** to score how well a track fits the conversation.



## Step 4 – write the reply

APPROACH

OPTIONAL

Can you find something with a bit more of that gritty, raw blues rock energy?



“...if you’re craving that gritty, raw blues rock edge, this one’s got it: Quaker City Night Hawks bring a smoky, soul-drenched intensity to ‘Good Evening’...”

---

A second LLM (**Qwen3-30B**) writes one short reply about the top track, from its details and the state.



# 5 How it scored

the official results

# How it scored

RESULTS CORE

| OFFICIAL CODABENCH SPLIT    | NDCG@20 | CAT.   | LEX.   | JUDGE | COMPOSITE |
|-----------------------------|---------|--------|--------|-------|-----------|
| Blind-A · development phase | 0.4380  | 0.0313 | 0.8028 | 4.70  | 0.5799    |
| Blind-B · final, 29th of 40 | 0.2537  | 0.0315 | 0.7862 | 3.30  | 0.3811    |

NDCG@20

**-42%**

0.4380 → 0.2537

JUDGE (1-5)

**-1.40**

4.70 → 3.30

DIVERSITY TERMS

**flat**

catalog and reply-wording  
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.

# Where our points came from

RESULTS

OPTIONAL

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

## OUR BLIND-B DECOMPOSITION

|                                |        |        |
|--------------------------------|--------|--------|
| $0.50 \times \text{nDCG@20}$   | 0.2537 | 0.1269 |
| $0.10 \times \text{cat. div.}$ | 0.0315 | 0.0032 |
| $0.10 \times \text{lex. div.}$ | 0.7862 | 0.0786 |
| $0.30 \times (3.30 - 1)/4$     |        | 0.1725 |
| Composite                      |        | 0.3811 |

## WHERE THE SCORE CAME FROM

The **judge term** out-earned **nDCG**, and category diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.



# 6 What we learned

four lessons, each with its evidence

# Where the failures were — all 80 Blind-B turns

ANALYSIS CORE

| DIAGNOSED GAP          | TURNS | MEANING                                                |
|------------------------|-------|--------------------------------------------------------|
| Ranking                | 31    | a better, non-violating candidate sat below our top-20 |
| Constraint not applied | 8     | an extracted filter never reached the pool             |
| State extraction       | 3     | the extractor missed something the user said           |
| Entity resolution      | 1     | a name never resolved to a catalog ID                  |

## TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

## HOW THIS WAS COUNTED

43 diagnoses across 32 distinct turns. A **rule-based check** of each turn's system trace — not an LLM, and not a relevance label.

Extraction is **3 of 43** diagnoses. The failure is overwhelmingly downstream of understanding the conversation.



# Extracted, then ignored

WENT WRONG

CORE



*"I gotta break away from **Deltron** and **Del** completely"*

**Dead Prez — Hip Hop**

state held both artists as avoid

IGNORED



*"branching out to **other** singer-songwriter artists"*

**Ryan Adams — When The Stars Go Blue**

state held Ryan Adams as avoid

IGNORED



*"still focusing on **Juanes**... haven't found the depth I'm after"*

**Juanes — Me Enamorado De Ti**

state held Juanes as rejected

IGNORED

Three sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway.**



# Asked for, served

WENT WRONG

CORE

1

*“the Kamelot track from **Silverthorn**”*

**Kamelot — Fallen Star**

from *Haven*. We extracted “Silverthorn” but never filtered on it.

CONSTRAINT

2

*“exactly **126.70 bpm**, in **G minor**”*

**no tempo or key in the catalog**

and we added no data from outside it

NO DATA

3

*“**‘Watercolors’** by **Pat Metheny**”*

**Pat Metheny — Alfie**

reply: “Got it — here’s ‘Alfie’...” — “Watercolors” is not in the catalog

RESPONSE



# What the training labels look like

WENT WRONG

CORE



*“I keep asking for **Arcade Fire** — that's the only song I want”*

label: **Arctic Monkeys — This House is a Circus**

DOES NOT



*“similar chill vibe, but from a different artist”*

label: **Bonobo — Jets**

MOVES



*“what **other artists** fit this darkwave style?”*

label: **Tropic of Cancer — The Dull Age**

MOVES



*“what **other** orchestral tracks would you recommend?”*

label: **Nick Phoenix — Kogan**

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**

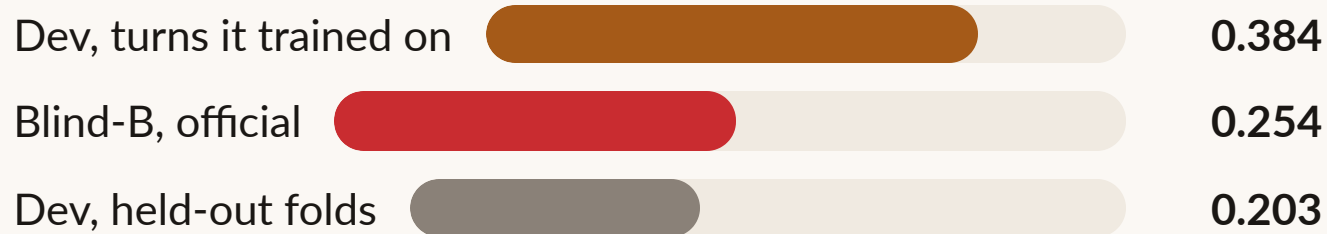
Why: the answer is the simulated recommender's pick from a 16–32-track pool, made without seeing the listener's goal.



# We tuned on what we measured

LESSON

OPTIONAL



## WHAT HAPPENED

The ranker was trained **and** checked on the same 8,000 dev turns. The number we steered by was a replay of turns it had already seen.

nDCG@20. The top row is an earlier model version, shown because it is the number we selected on. Source: paper, Table 3 and §5.3.



# Four ways the reply went wrong

RESPONSE

OPTIONAL

1

**“Got it — here's *Alfie*...”**

for a track we do not have

CLAIMS COMPLIANCE

2

**“Sorry, ‘Cthulhu Dawn’... doesn’t match your request...”**

about our own rank-1 pick

ARGUES AGAINST ITSELF

3

**Asked a question, given a recommendation**

“is it the rhythm or the instrumentation?”

DODGES

4

**“hauntingly beautiful... raw emotion”**

no evidence from the track or the conversation

GENERIC

Row 2 follows the prompt’s instruction **not to oversell** a weak match — so the reply argues against our own top pick.



# Everything is released

THANK YOU

CORE



Paper, poster, slides, code, models, data

[npatta01.github.io/music-crs-2026](https://npatta01.github.io/music-crs-2026)

# → Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

- |                    |                                          |                    |                                                      |                    |                                              |
|--------------------|------------------------------------------|--------------------|------------------------------------------------------|--------------------|----------------------------------------------|
| <a href="#">1</a>  | The task                                 | <a href="#">2</a>  | Glossary                                             | <a href="#">3</a>  | State extraction                             |
| <a href="#">4</a>  | The state schema                         | <a href="#">5</a>  | State on a lookup turn                               | <a href="#">6</a>  | State on a request the catalog cannot answer |
| <a href="#">7</a>  | Six request types, six shapes of state   | <a href="#">8</a>  | Routing flags decide which branches fire             | <a href="#">9</a>  | Entity resolution                            |
| <a href="#">10</a> | The eleven branches                      | <a href="#">11</a> | One index                                            | <a href="#">12</a> | Fusion and fallback                          |
| <a href="#">13</a> | The b1 bi-encoder                        | <a href="#">14</a> | b1 training                                          | <a href="#">15</a> | Noisy-label down-weighting                   |
| <a href="#">16</a> | Response generation                      | <a href="#">17</a> | Development-split results                            | <a href="#">18</a> | What is not ideal                            |
| <a href="#">19</a> | Failure 2 — nothing in the data to match | <a href="#">20</a> | Failure 3 — the better track was already in the pool | <a href="#">21</a> | Label-free audit                             |
| <a href="#">22</a> | The target often ignores the request     | <a href="#">23</a> | How the relabelling worked                           | <a href="#">24</a> | What the relabelling found                   |
| <a href="#">25</a> | Limits of our own analysis               | <a href="#">26</a> | What we would do differently                         | <a href="#">27</a> | Related work                                 |
| <a href="#">28</a> | Reproducing the run                      |                    |                                                      |                    |                                              |

## EVERY ASSISTANT TURN

- Retrieve **20 tracks** from a shared **47,071-track** catalog
- Write a short reply presenting the top recommendation

## PER-TRACK METADATA

Title, artist, album, a median of **17** genre/mood tags, popularity, release date, plus precomputed text, audio, cover-art and collaborative-filtering embeddings.

## SPLITS

|                   |                                  |
|-------------------|----------------------------------|
| Train             | 15,199 sessions (~121.6k turns)  |
| Development       | 1,000 sessions × 8 turns = 8,000 |
| Blind-A / Blind-B | 80 sessions each, final turns    |

Blind-B additionally withholds the conversation-goal annotations present in training data.

## BUILT ON

TalkPlayData 2. The conversations are **LLM-generated**, not observed listening sessions — which matters later.



# Glossary

BACKGROUND

|                              |                                                                                                                      |                                      |                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Typed state</b>           | Structured fields an LLM reads off the conversation each turn: artists to anchor on, rejected tracks, filters, mood. | <b>BM25</b>                          | A classic keyword-search score over text fields.                                                         |
| <b>Dense embedding</b>       | A vector from a neural model; texts with similar meaning land close together.                                        | <b>CLAP</b>                          | A model that puts text and audio in one space, so a description can find a sound.                        |
| <b>SigLIP-2</b>              | The same idea for text and images — here, album covers.                                                              | <b>Collaborative filtering (BPR)</b> | Vectors learned from who listened to what; BPR is the training method.                                   |
| <b>Centroid search</b>       | Average the vectors of tracks the listener liked, then search near that average.                                     | <b>RRF</b>                           | Reciprocal-rank fusion: merge ranked lists by summing $1 / (k + \text{rank})$ per track.                 |
| <b>LambdaMART · LightGBM</b> | Gradient-boosted decision trees trained to order a list; LightGBM is the library.                                    | <b>Bi-encoder</b>                    | One model encodes the conversation and the track separately; relevance is the cosine of the two vectors. |
| <b>LLM judge</b>             | An LLM that rates the reply 1–5; the organisers run it.                                                              | <b>In-sample vs out-of-fold</b>      | Scoring on turns the model trained on, versus on held-out folds. In-sample overstates.                   |



# State extraction

APPROACH

## THE EXTRACTOR

DeepSeek-V4-Flash (build 0423), schema-constrained, given the dialog so far and the IDs of tracks already played.

## DELIBERATELY BLIND

No conversation goals, goal progress, or user profiles reach the extractor. The state is whatever an LLM can read off the conversation.

## WHY THIS MODEL

Cheaper extractors under-filled the optional fields — rejections and facets especially — so every final run uses this one. Per-turn extraction is a paid API call, which is what kept us off the training split.

## PROMPT VERSION

ConversationStateV1, projected to V0Plus for the downstream contract.



# What the state can hold

APPROACH

## THE REQUEST

- request type
- intent mode
- routing flags

## WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

## WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

## WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

## WHAT IT IS ABOUT

- lyrical theme

## BOUNDS

- release-year window
- process constraints



# State on a lookup turn

STATE

**User:** “Can you tell me which song from Peter Doherty's *Grace/Wastelands* was frequently mentioned as one of the best tracks of 2009?”

**Extracted state:**

```
request_type: exact_album
intent_mode: refinement
artist: Peter Doherty (must_use)
album: Grace/Wastelands (must_use)
facet: "best tracks of 2009"
routing: exact_entity_probe = true
```

**We served:** Peter Doherty – “Arcady”

## THIS ONE WORKS

Two hard entities plus a soft facet. The probe flag opens the catalog lookups, the album constrains the pool, and *Arcady* is genuinely from *Grace/Wastelands*.

## WHY IT IS EASY

Everything the listener asked for is a field the catalog actually has. Compare the next failures, where it is not.



# State on a request the pipeline cannot use

STATE

**User:** “I'm looking for an album with a really striking, abstract cover art. It had some bold colors and shapes, almost like a modern painting.”

**Extracted state:**

```
request_type: attribute_search
facet: striking abstract cover art
facet: bold colors and shapes
facet: like a modern painting
routing: image_or_visual_search = true
```

**We served:** Frank Ocean – “Pink + White”

## EXTRACTION: RIGHT

Three visual facets and the image flag — exactly what the schema is for.

## PIPELINE: NOWHERE TO PUT IT

The flag opens the gated SigLIP-2 cover-art branch, but **the reranker has no features for that branch**. Whatever it returns cannot be weighed against anything else.

---

This is the gap in miniature: the state understood the request, and the rest of the pipeline had no way to act on it.



# Six request types, six shapes of state

APPROACH

| REQUEST TYPE                  | A REAL TURN                                                         | WHAT THE EXTRACTOR PULLED OUT                                                                    |
|-------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>exact_track</code>      | "I wanted <i>Watercolors</i> by Pat Metheny specifically."          | track = <i>Watercolors</i> <b>must_use</b> · artist = Pat Metheny <b>must_use</b>                |
| <code>exact_album</code>      | "Which song from Peter Doherty's <i>Grace/Wastelands...</i> "       | artist + album <b>must_use</b> · facet "best tracks of 2009"                                     |
| <code>same_artist</code>      | "Got another track known for his guitar work?"                      | artist <b>must_use</b> · prior track <b>do_not_use</b> · facets: guitar work, melodic complexity |
| <code>new_artist</code>       | "...branching out to other singer-songwriter artists."              | artist = Ryan Adams & The Cardinals <b>do_not_use</b> · facets: acoustic, introspective          |
| <code>attribute_search</code> | "...a really striking, abstract cover art, bold colors and shapes." | three visual facets · routing flag <b>image_or_visual_search</b>                                 |
| <code>hidden_target</code>    | "A Lil Wayne song from an experimental phase."                      | artist <b>must_use</b> · facet "experimental" · flag <b>hidden_target_search</b>                 |

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.



# Routing flags decide which branches fire

APPROACH

| FLAG                   | OPENS                                    |
|------------------------|------------------------------------------|
| exact_entity_probe     | catalog lookups — discography, album     |
| lyric_search           | the Qwen3 lyrics view                    |
| feature_articulation   | CLAP text→audio on the sonic description |
| image_or_visual_search | the gated SigLIP-2 cover-art branch      |
| hidden_target_search   | widens the pool; suppresses exact pins   |

## A REAL VISUAL REQUEST

“I’m looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

## AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.



# Grounding names to the catalog

APPROACH

## NAMES → IDS

Fuzzy string matching of artist and track names against the catalog name lists, keeping matches above **80/100**. Downstream stages only ever see identifiers.

## FREE TEXT → TAGS

- 1 Exact match against the tag vocabulary
- 2 Alias match
- 3 Substring match
- 4 Embedding nearest neighbour, cosine  $\geq 0.6$ , top 3

Used by the BM25 branch. Outside it, tag matching falls back to exact strings — part of why so little of the state became executable.

# The eleven retrieval branches

APPROACH

| BRANCH               | SPACE            | QUERY                       |
|----------------------|------------------|-----------------------------|
| BM25 (fielded)       | lexical          | state text + resolved tags  |
| Dense text × 3       | Qwen3 views      | state-built strings         |
| CLAP text→audio      | audio            | sonic description           |
| Anchor centroids × 3 | SigLIP-2/CLAP/CF | liked-track centroids †     |
| User-CF centroid     | CF-BPR           | the listener's own vector   |
| Lookups × 2          | catalog          | discography; era popularity |

## GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

## POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.

Appendix

B — back to the talk · A — index · number + Enter — jump

# One LanceDB collection

APPROACH

| COLUMN                  | MODEL                    | DIM  |
|-------------------------|--------------------------|------|
| Lyrics text             | Qwen3-Embedding-0.6B     | 1024 |
| Metadata text           | Qwen3-Embedding-8B       | —    |
| Attributes text         | Qwen3-Embedding-8B       | —    |
| Audio                   | LAION-CLAP               | 512  |
| Cover art               | SigLIP-2                 | 768  |
| Collaborative filtering | BPR                      | 128  |
| Bi-encoder document     | b1 (fine-tuned Qwen3-4B) | 2560 |

## THE INTENT

Express every filter in one space, so retrieval could in principle collapse into a single composed query.

## WHAT HAPPENED

We never got there. Branches stayed separate and most state fields never became query predicates at all.



# Weighted RRF — baseline and fallback

APPROACH

## FUSION

Weighted reciprocal-rank fusion,  $k = 60$ , top-1000 across the branch pools.

## POST-FUSION ADJUSTMENTS

- Drop rejected artists and tracks
- Drop tracks already played
- Promote / demote on exact tag membership
- Year-range decay

## AS A BASELINE

**0.1492**

devset nDCG@20, no training

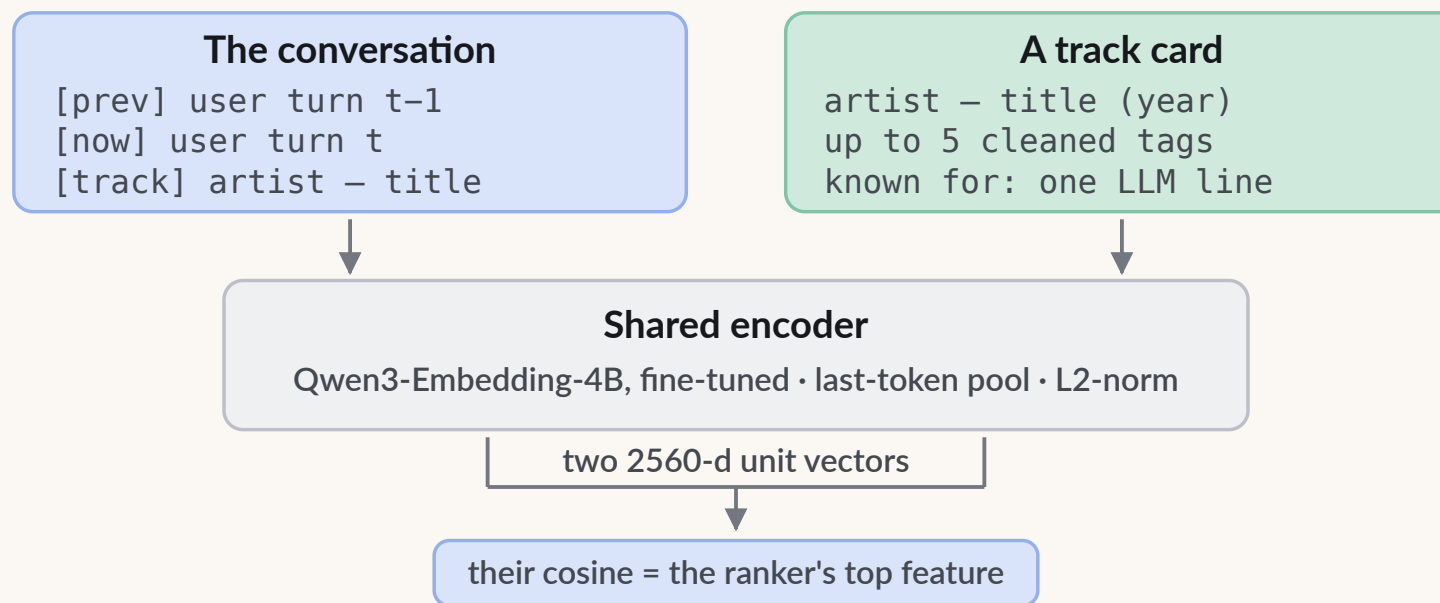
## STILL THE FALLBACK

If the learned ranker is unavailable, this order ships. It is the only part of the ranking stack with no trained component.



# The bi-encoder behind that top feature

APPROACH





# How b1 was trained

APPROACH

$$L = -(1/B) \sum_i \log \left[ \exp(v_{q_i}^T v_{d_{i+}} / \tau) / \sum_{d \text{ in } D_i} \exp(v_{q_i}^T v_d / \tau) \right] \quad \tau = 1/20$$

## POSITIVES

**53,885**

(turn, track) pairs whose *next* turn's goal-progress says the recommendation moved the session forward

## NEGATIVES

**4 + batch**

four mined hard negatives per positive, plus every in-batch document

## RETRIEVAL VS RANKING

As a twelfth *retrieval* branch it added little to the union's coverage and was dropped. As a *ranking* feature it dominates the fit — out-of-fold nDCG@20 0.1970 → 0.2032.

Not isolated: which parts of the rendering matter, and whether the effect carries to the blind splits.



# Down-weighting contradicted turns

APPROACH

## WEIGHT, BY WHAT THE NEXT TURN SAYS

|                                    |            |
|------------------------------------|------------|
| User rejects the labelled track    | × 0.3      |
| Goal-progress says it did not help | × 0.3      |
| User rejects only its artist       | × 0.6      |
| <b>Floor</b>                       | <b>0.2</b> |

Chosen by hand. Never tuned.

## THE GOAL-ANNOTATION ASYMMETRY

Goal-progress annotations shape **training labels** — these weights, and the bi-encoder's positives — but are never **features**. Blind-B removes them, so the deployed ranker is goal-free at inference.

# Response generation – 30% of the composite

APPROACH

## THE DEPLOYED STYLE INSTRUCTION

“Write 1–2 concise sentences about only the selected track. Prioritize the latest user request and extracted state over older conversation history. If the track is reasonably aligned, explain the fit with one specific supported reason. If it clearly conflicts with an explicit avoid/new-artist constraint, do not oversell it or blame the system; briefly frame the limitation and the closest supported reason.”

## SETUP

Qwen3-30B-A3B-Instruct-2507, temperature 0. One pass over the top-1 track as a delimited XML block ( $\leq 10$  tags) plus the **latest** state.

## HOW THE PROMPT WAS CHOSEN

Eight variants swept over a frozen Blind-A retrieval output → judge **4.70**, no recommendations changed. Selected on the leaderboard score.

## MISSING

No multi-draft sampling, no selection, no checking, no repair pass.



# Development split — what we steered on

RESULTS

| SYSTEM (HOW MEASURED)                       | NDCG@20       |
|---------------------------------------------|---------------|
| Weighted RRF fusion (no training)           | 0.1492        |
| LambdaMART, out-of-fold CV                  | 0.1970        |
| + b1_cos (deployed feature set)             | 0.2032        |
| Historical v10 LambdaMART, <b>in-sample</b> | 0.3844        |
| <b>Official Blind-B</b>                     | <b>0.2537</b> |

### DO NOT OVER-READ THE FIRST GAP

The fusion and learned-ranker configurations used **different candidate-pool depths**, so 0.149 vs 0.197–0.203 does not isolate the effect of learned re-ranking.

### THE IN-SAMPLE ROW

A historical goal-aware model without b1\_cos.  
Not directly comparable — shown because it is the number we were selecting on.



# What is not ideal

WENT WRONG

1

## We tuned on what we measured

The development set was both our training and our evaluation data.

2

## We extracted more than we used

Most fields in the state never reached retrieval or ranking at all.

3

## Nothing from outside the catalog

No tempo, key or soundtrack. Past one CF vector, no listening behaviour.

4

## The reply was never checked

Asked for a track we did not have, it said “Got it” and offered another.



## Failure 2 — nothing in the data to match

ANALYSIS

|           |                                                                                         |
|-----------|-----------------------------------------------------------------------------------------|
| USER      | <i>"I wanted Watercolors by Pat Metheny specifically. Can you play that one?"</i>       |
| STATE     | track <b>Watercolors</b> must_use · artist <b>Pat Metheny</b> must_use — read perfectly |
| REALITY   | the track is not in the 47,071-track catalog                                            |
| WE SERVED | <b>Pat Metheny — Alfie, with the reply "Got it — here's 'Alfie'..."</b>                 |
| USER      | <i>"Exactly 126.70 bpm, in G minor."</i>                                                |
| REALITY   | no tempo field, no key field — no branch can match on either                            |
| USER      | <i>"A song from the Breaking Bad soundtrack."</i>                                       |
| REALITY   | no film or TV metadata anywhere in the catalog                                          |

Extraction was correct every time. The gap is **knowledge, not understanding** — and when the answer does not exist, the system should say so rather than offer the nearest thing in its place.



# Failure 3 — the better track was already in our pool

ANALYSIS

| WHAT THEY ASKED FOR                                            | WE SERVED AT RANK 1               | ALREADY IN OUR POOL                                    | POOL RANK |
|----------------------------------------------------------------|-----------------------------------|--------------------------------------------------------|-----------|
| nu-jazz, downtempo, female vocals —<br><i>not</i> Parov Stelar | Parov Stelar — Booty Swing        | St Germain — What You Think About...                   | 34        |
| new underground hip-hop, not Deltron                           | Dead Prez — Hip Hop               | Souls Of Mischief — Make Your Mind Up                  | 31        |
| hardcore, not classic hip-hop                                  | Dead Prez — Hip Hop               | Papa Roach — Getting Away With Murder                  | 49        |
| that carol, but not Beegie Adair again                         | Beegie Adair — The Christmas Song | Frank Sinatra — Have Yourself A Merry Little Christmas | 63        |

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.



# Auditing the submitted run

ANALYSIS

## SINGLE-JUDGE QUALITY RATING

78%

of Blind-B turns rated weak-or-bad by one DeepSeek-V4-Flash judge

## READ AS

Diagnostic, not ground truth. Blind-B exposes no relevance labels, so only the validation failure is actually *measured*.

## DETERMINISTIC TRACE AUDIT — 43 DIAGNOSES OVER 32 TURNS

|                                    |    |
|------------------------------------|----|
| Heuristic alternative below top-20 | 31 |
| Under-applied constraint           | 8  |
| State-extraction miss              | 3  |
| Entity-resolution miss             | 1  |

Categories overlap. The alternative was chosen by positive-term metadata matches plus popularity — an audit heuristic, not a relevance judgment.



# The target often ignores the request

ANALYSIS

**User:** “...something with a similar chill, electronic vibe, but **from a different artist?**” (just played: Bonobo)

**Target:** **Bonobo – “Jets”** · annotated MOVES toward goal

**Next turn:** “‘Jets’ is cool, but I was actually hoping for something **from a different artist this time...**”

**Target:** **Bonobo – “Pieces”** · annotated MOVES toward goal

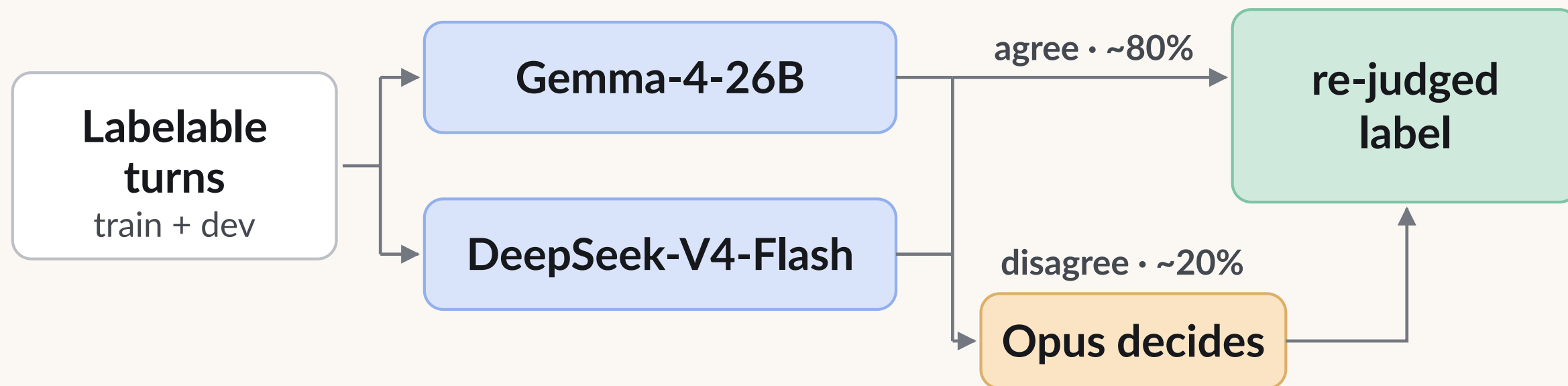
Ground truth is simply **whatever played next**, in a session an LLM wrote. Our own model anchors on the just-played artist — we think it learns that here.



# How the LLM relabelling worked

ANALYSIS

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



# What the relabelling found

ANALYSIS

## RE-JUDGED

**113,393**

labelable train + dev turns

## NOT SETTLED AS A GOOD FIT

**61%**

of the 106,393 training turns

## OUTRIGHT ANCHORING

**18,222**

17% — the label keeps the rejected artist

## WHAT HAPPENED WHEN WE USED THEM

A ranker trained on the re-judged labels scored **lower** — so the submission kept the originals.

## THE CATCH

That score is computed **against the original labels**, so it could not have favoured the relabeling. We release the re-judged labels as evidence to audit, not as a corrected benchmark.



# Limits of our own analysis

ANALYSIS

- The relabeling was **never checked against human judgments**.
- The Blind-B quality audit used a **single** LLM judge.
- The trace audit was **heuristic**, not a relevance judgment.
- The conversations are **LLM-generated**, not observed listening sessions.
- Neither blind split exposes relevance labels — blind-set failure rates and relabeling rates are **diagnostic, not ground truth**.
- Development-split and leaderboard numbers are **separate measurements**.



# What we would do differently

CONCLUSION

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

## → Related work — four established lines

A1

### CONVERSATIONAL STATE TRACKING

CRS systems commonly track dialog state explicitly and condition retrieval on it. We follow that, but compile the state with a **prompted LLM** instead of a trained tracker.

### MODALITY-SPECIFIC MUSIC ENCODERS

Contrastive language-audio pretraining (CLAP), vision-language over cover art (SigLIP-2), collaborative filtering (BPR), general text embeddings (Qwen3).

### RETRIEVE-THEN-RERANK AT CATALOG SCALE

Pool multiple sources with reciprocal-rank fusion; rank with LambdaMART.

### OUR CLAIM

What we report is **how that combination behaved** — not a new technique.



# Reproducing the run

A2

## FROZEN REPLAY – NO CREDENTIALS

Source the offline bundle's env and every LLM call is served from a curated cache. A miss hard-fails rather than silently spending money.

```
source .repro/scripts/activate_repro_env.sh
```

## LIVE RE-RUN

```
python run_experiment.py --backend modal --tid state_ranker_v10_lgbm_blindset_B --eval_dataset blindset_B
```

## ARTIFACTS

Models and data: [hf.co/datasets/Npatta01/music-crs-repro-2026](https://huggingface.co/datasets/Npatta01/music-crs-repro-2026)